AI-GENERATED DIAGNOSTIC REPORT

Comprehensive Clinical Analysis

Brain MRI · Cardiac ECG · Cardiovascular Risk Assessment

PATIENT NAME	REPORT ID	DATE GENERATED
Mohammed Panchla	9B0B74A9	20 February 2026, 14:56
AGE / GENDER	**MODELS USED**	**OVERALL STATUS**
18 yrs · Male	5 AI Models	CRITICAL

CLINICAL REPORT SCOPE

Neuro Imaging
Tumor detection classification with multi-class probability output.

Region Segmentation
Pixel-wise mask and overlay estimation for lesion localization support.

Cardio Assessment
ECG anomaly prediction and cardiovascular risk estimation summary.

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Executive Summary

Integrated findings across all diagnostic modules

CRITICAL

Critical findings detected. Immediate specialist consultation recommended.

BRAIN TUMOR DETECTION

Glioma

HIGH severity · 100.0% conf.

CARDIAC ECG

ABNORMAL

100.0% confidence

CARDIOVASCULAR RISK

High Risk

Stroke: High Risk

PATIENT CLINICAL PARAMETERS

PARAMETER	VALUE	PARAMETER	VALUE
Resting Blood Pressure	160.0 mmHg	BMI	45.0
Cholesterol	310.0 mg/dL	Avg. Glucose Level	256.0 mg/dL
Max Heart Rate	95.0 bpm	Fasting Blood Sugar	No (≤120)
Hypertension	Yes	Smoking Status	Smokes

Brain Tumor Detection

VGG16 Transfer Learning · BraTS Dataset · 4-Class Classification

Test Acc: 93.25% | Val: 98.48% | VGG16 · 4 Classes

DIAGNOSIS

Glioma Tumor

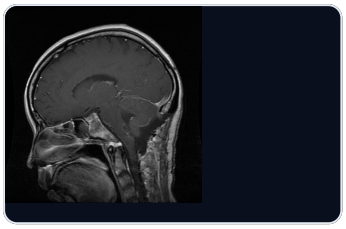
Glioma is an aggressive tumour originating in glial cells. Immediate referral to a neuro-oncologist is strongly recommended for further imaging and biopsy evaluation.

100.0%
CONFIDENCE

SCAN PREVIEW · FILE INFORMATION

▲ ABNORMALITY DETECTED

FILE FORMAT
JPG



MODEL INPUT
224 × 224

ANALYSIS TIME
~0.8s

ARCHITECTURE
VGG16

CLASSES
4

NORMALISE
ImageNet

PER-CLASS PROBABILITY BREAKDOWN

SOFTMAX OUTPUT

GLIOMA	100.0%
MENINGIOMA	0.0%
NOTUMOR	0.0%
PITUITARY	0.0%

TUMOR CLASSES · RISK REFERENCE

Glioma	HIGH RISK
Meningioma	MODERATE
Pituitary	MONITOR
No Tumor	NORMAL

DETECTION CONCLUSION

Primary MRI classification indicates **Glioma** with confidence **100.0%**. Correlate with segmentation and full clinical context.

Tumor Segmentation Analysis

U-Net + VGG16 Encoder · LGG Kaggle Dataset · Binary Segmentation

Val IoU:	Val Dice:	Test IoU:	Test Dice:
71.1%	78.2%	73.9%	81.2%

SEGMENTATION DIAGNOSIS

Tumor Detected

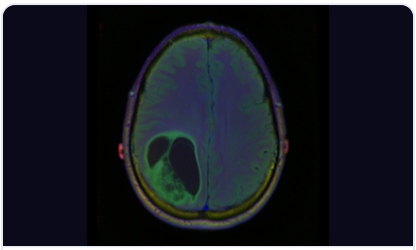
TUMOR COVERAGE

5.97%

TUMOR COVERAGE	TUMOR PIXELS	PARAMETERS
5.97%	3913	23.7M
of total scan area	/ 65,536 total px	U-Net + VGG16

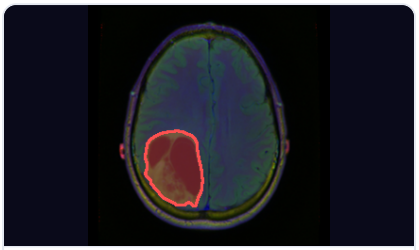
Tumor Pixels **3913 / 65,536 px**

SEGMENTATION OUTPUT — MRI · MASK · OVERLAY



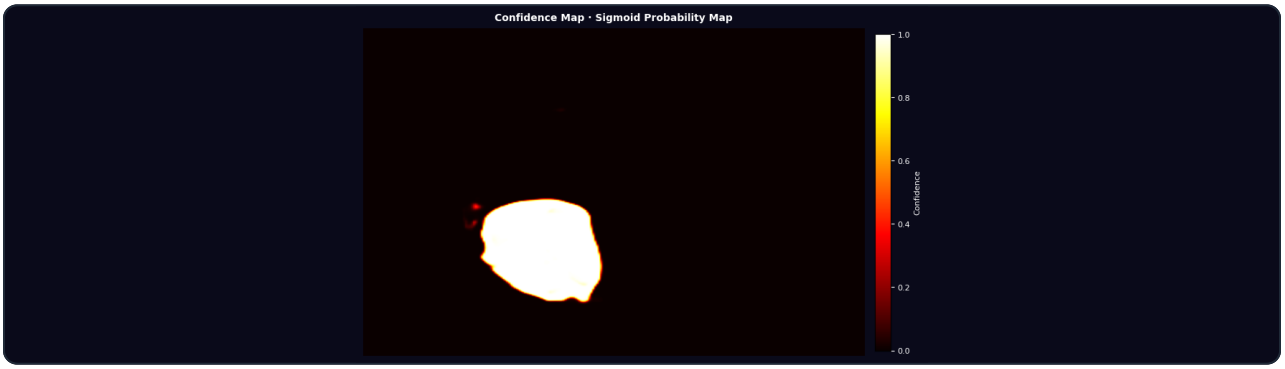
MRI SCAN

TUMOR MASK



OVERLAY

MODEL CONFIDENCE HEATMAP



MODEL PERFORMANCE

VAL IOU	VAL DICE
71.1%	**78.2%**
Segmentation	Segmentation

TEST IOU	TEST DICE
73.9%	**81.2%**
Generalisation	Generalisation

> Segmentation model: Val IoU 71.1%, Dice 78.2% (LGG dataset) — Coverage and mask estimates are subject to model confidence threshold (sigmoid > 0.5).

SEGMENTATION CONCLUSION

Segmentation indicates estimated tumor coverage of **5.97%** (3913 pixels). Review MRI scan, tumor mask and overlay together for clinical interpretation.

Cardiac ECG Analysis

CNN + LSTM · PTB Diagnostic ECG Dataset · Binary Classification

• F1: 99.46%

• AUC: 99.99%

CNN + LSTM

ABNORMAL HEARTBEAT

Cardiac abnormality detected — specialist review strongly recommended

100.0%

CONFIDENCE

ECG WAVEFORM · 187 TIME STEPS

ABNORMAL · P(ABNORMAL): 100.0%

PROBABILITY DISTRIBUTION

PTB Dataset · Normal / Abnormal

Abnormal	100.0%	100.0%
Normal		0.0%

CLASSIFICATION	ABNORMALITY PROB.	SIGNAL LENGTH	DECISION THRESHOLD
ABNORMAL	100.0%	187 steps	0.50

HOW IT WORKS

Step 1	Parse 187 time steps from CSV input
Step 2	Conv1D blocks — local feature extraction (64→128→256)
Step 3	LSTM (128 units) — temporal pattern modelling
Step 4	Sigmoid → probability output
Step 5	Threshold at 0.50 → Normal / Abnormal

ECG CONCLUSION

ECG analysis indicates **ABNORMAL** rhythm with **100.0%** confidence. Interpret in conjunction with cardiovascular risk metrics.

Cardiovascular Risk & Summary

Logistic Regression · Heart Disease Prediction · Stroke Risk Assessment

HEART DISEASE RISK

93.8%

HIGH RISK

STROKE RISK

7.9%

HIGH RISK

CLINICAL INPUT PARAMETERS USED FOR PREDICTION

PARAMETER	VALUE	PARAMETER	VALUE
Age	18 years	BMI	45.0
Resting Blood Pressure	160.0 mmHg	Avg. Glucose Level	256.0 mg/dL
Cholesterol	310.0 mg/dL	Max Heart Rate	95.0 bpm
Hypertension	Yes	Smoking Status	Smokes

Key Findings

MODULE	FINDING	SEVERITY / RISK
Brain MRI — Detection	Glioma · 100.0% confidence	HIGH
Brain MRI — Segmentation	Coverage 5.97% · 3913 pixels	ESTIMATED
Cardiac ECG	ABNORMAL heartbeat · 100.0% confidence	ATTENTION
Heart Disease Risk	93.8% probability	HIGH
Stroke Risk	7.9% probability (threshold 0.0202)	HIGH

CARDIOVASCULAR CONCLUSION

Combined profile indicates heart risk **93.8%** and stroke risk **7.9%**. Clinical follow-up and physician-led management are recommended according to risk band.

Model Performance & Compliance

Reference metrics and legal note

AI MODEL PERFORMANCE REFERENCE

MODEL	ARCHITECTURE	DATASET	PERFORMANCE
Tumor Detection	VGG16	BraTS	Test Acc: 93.25% · Val: 98.48%
Tumor Segmentation	U-Net + VGG16	LGG Kaggle	Val IoU: 71.1% · Dice: 78.2%
ECG Classification	CNN + LSTM	PTB ECG	F1: 99.46% · AUC: 99.99%
Heart Disease	Logistic Regression	UCI Heart	~85% accuracy
Stroke Risk	Logistic Regression + SMOTE	Kaggle Stroke	Threshold: 0.30

⚠ Important Legal Disclaimer

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